

Finite-Prime Toeplitz Dynamics, KMS States, and Euler Factors

Steven Sabean
steven@ceadmth.org

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Abstract

The Bost–Connes system is a paradigmatic example of an arithmetic C^* -dynamical system whose equilibrium structure is governed by the Riemann zeta function. In this paper we isolate and analyze an elementary finite-prime model that retains the Euler-product mechanism while avoiding the additional symmetry and adelic structure of the full Bost–Connes system.

Let S be a finite set of rational primes and let

$$\Lambda_S = \left\{ \prod_{p \in S} p^{k_p} : k_p \in \mathbb{N} \right\}.$$

On $\ell^2(\Lambda_S)$ we consider the multiplicative shifts $V_m e_n = e_{mn}$, the Toeplitz algebra $\mathcal{T}_S = C^*(1, V_p : p \in S)$, and the dynamics $\alpha_t(V_m) = m^{it} V_m$. For every real $\beta > 0$ we prove that $(\mathcal{T}_S, \alpha)$ has a unique KMS_β state, given on the standard monomials by

$$\varphi_{\beta, S}(V_m V_n^*) = \delta_{m, n} m^{-\beta}.$$

Its Gibbs partition function is

$$Z_S^{\text{fin}}(\beta) = \sum_{n \in \Lambda_S} n^{-\beta} = \prod_{p \in S} (1 - p^{-\beta})^{-1}.$$

We then recover the archimedean factor

$$Z_\infty(\beta) = \pi^{-\beta/2} \Gamma\left(\frac{\beta}{2}\right)$$

from the Mellin transform of the Gaussian. The resulting product

$$Z_S(\beta) = \pi^{-\beta/2} \Gamma\left(\frac{\beta}{2}\right) \prod_{p \in S} (1 - p^{-\beta})^{-1}$$

is a finite-prime truncation of the completed zeta factor. The finite-prime factor is a genuine trace-class Gibbs partition function; the archimedean factor is a Mellin normalization, and we keep these two constructions conceptually distinct.

1 Introduction

The interaction between operator algebras, equilibrium statistical mechanics, and arithmetic became particularly striking in the work of Bost and Connes [1]. Their C^* -dynamical system has the Riemann zeta function as its partition function and exhibits a nontrivial KMS phase structure tied to arithmetic symmetry. The present paper studies a much simpler object. We retain only finitely

many prime modes and work directly with the Toeplitz shifts generated by multiplication by those primes.

The basic observation is elementary but structurally suggestive. For a prime p and real $\beta > 0$,

$$\sum_{k=0}^{\infty} e^{-\beta k \log p} = \sum_{k=0}^{\infty} p^{-\beta k} = (1 - p^{-\beta})^{-1}.$$

Thus the local Euler factor is the Gibbs partition function of a single occupation-number mode with energy spacing $\log p$. For a finite set of primes the unique-factorization theorem makes the modes independent: if

$$n = \prod_{p \in S} p^{k_p},$$

then

$$\log n = \sum_{p \in S} k_p \log p.$$

The finite Euler product is therefore expected to arise from a diagonal Hamiltonian with energy $\log n$.

Our aim is to make that statement precise at the level of a C^* -dynamical system. Let S be a finite set of rational primes and define

$$\Lambda_S := \left\{ \prod_{p \in S} p^{k_p} : k_p \in \mathbb{N} \right\}, \quad (1)$$

where $\mathbb{N} = \{0, 1, 2, \dots\}$. Thus $1 \in \Lambda_S$, and if $S = \emptyset$ then $\Lambda_S = \{1\}$. We represent this semigroup by multiplicative shifts on $\ell^2(\Lambda_S)$, form the Toeplitz algebra generated by the prime shifts, and equip it with the dynamics determined by the arithmetic energy $\log n$.

The main result is that for every real inverse temperature $\beta > 0$ this finite-prime system has a unique KMS state, with

$$\varphi_{\beta, S}(V_m V_n^*) = \begin{cases} m^{-\beta}, & m = n, \\ 0, & m \neq n, \end{cases}$$

and partition function

$$Z_S^{\text{fin}}(\beta) = \prod_{p \in S} (1 - p^{-\beta})^{-1}.$$

For fixed finite S , this Euler product is holomorphic in the larger half-plane $\Re(s) > 0$. The familiar restriction $\Re(s) > 1$ appears only when the finite set S is allowed to exhaust all rational primes, in which case the Euler products converge to $\zeta(s)$.

We also include the real place. The Mellin transform of the Gaussian gives

$$2 \int_0^{\infty} x^{s-1} e^{-\pi x^2} dx = \pi^{-s/2} \Gamma\left(\frac{s}{2}\right), \quad \Re(s) > 0,$$

the standard archimedean factor familiar from the adelic treatment of zeta and L -functions; see, for example, Tate's thesis [6]. Combining this Mellin normalization with the finite-prime Gibbs factor gives a finite truncation of the completed zeta factor.

The model studied here is inspired by the arithmetic quantum statistical mechanics of Bost–Connes, but it is not asserted to be a finite-stage subsystem, quotient, or crossed-product presentation of the full Bost–Connes algebra. Its purpose is narrower: to isolate the finite Euler factors in a completely explicit Toeplitz–KMS system for which existence, uniqueness, and the partition function can all be proved directly.

2 C*-dynamical systems and the KMS condition

We briefly recall the operator-algebraic language used below. Standard references for C*-dynamical systems and KMS states include Bratteli–Robinson [2]; for KMS states on Toeplitz and Nica–Toeplitz algebras, see also [4, 5].

A C*-dynamical system is a pair (A, α) , where A is a C*-algebra and

$$\alpha : \mathbb{R} \longrightarrow \text{Aut}(A)$$

is a point-norm continuous one-parameter group:

$$\alpha_{t+u} = \alpha_t \circ \alpha_u, \quad \|\alpha_t(a) - a\| \longrightarrow 0 \quad (t \rightarrow 0)$$

for every $a \in A$.

An element $a \in A$ is entire analytic for α if the map $t \mapsto \alpha_t(a)$ extends to an entire A -valued function $z \mapsto \alpha_z(a)$ on $\mathbb{C}$. Entire analytic elements form a dense *-subalgebra. For a real inverse temperature $\beta > 0$, a state φ is a KMS $_\beta$ state if

$$\varphi(ab) = \varphi(b\alpha_{i\beta}(a)) \tag{2}$$

on a dense *-algebra of entire analytic elements. This analytic-element form of the KMS condition is equivalent to the standard strip formulation under the usual continuity hypotheses [2].

When the dynamics is implemented on a Hilbert space by a self-adjoint Hamiltonian H ,

$$\alpha_t(a) = e^{itH} a e^{-itH},$$

and $e^{-\beta H}$ is trace class, the normalized Gibbs functional

$$\varphi_\beta(a) = \frac{\text{Tr}(e^{-\beta H} a)}{\text{Tr}(e^{-\beta H})}$$

is the canonical KMS state. The finite-prime model below is an infinite-dimensional but completely explicit instance of this mechanism.

3 The finite-prime Toeplitz system

Let

$$\mathcal{H}_S := \ell^2(\Lambda_S),$$

with orthonormal basis $\{e_n : n \in \Lambda_S\}$. For $m \in \Lambda_S$, define an isometry

$$V_m e_n := e_{mn}.$$

Then

$$V_m V_n = V_{mn}, \quad V_1 = 1, \quad V_m^* V_m = 1.$$

Unique factorization gives a canonical identification

$$\Lambda_S \cong \prod_{p \in S} \mathbb{N}, \quad (k_p)_{p \in S} \longmapsto \prod_{p \in S} p^{k_p}.$$

Consequently, if $m = \prod_{p \in S} p^{k_p}$, then

$$V_m = \prod_{p \in S} V_p^{k_p}.$$

Thus the prime shifts already generate all semigroup shifts. Let

$$\mathcal{T}_S := C^*(1, V_p : p \in S) \subset B(\mathcal{H}_S).$$

For a single prime this is the classical Toeplitz algebra generated by a proper isometry, in the sense of Coburn [3]. For general finite S , it is the concrete Toeplitz algebra of the free abelian semigroup $\Lambda_S \cong \mathbb{N}^{|S|}$, a particularly elementary example of the multivariable Toeplitz constructions that appear in the modern KMS literature [5]. With this convention, if $S = \emptyset$, then $\mathcal{H}_S = \mathbb{C}$ and $\mathcal{T}_S = \mathbb{C}$. The linear span of monomials

$$V_m V_n^*, \quad m, n \in \Lambda_S,$$

is a dense $*$ -subalgebra of $\mathcal{T}_S$.

Rather than beginning with an unbounded Hamiltonian, define the dynamics directly by the diagonal unitaries

$$U_t e_n := n^{it} e_n = e^{it \log n} e_n, \quad t \in \mathbb{R}. \quad (3)$$

Since $|n^{it}| = 1$, each U_t is unitary, and

$$U_0 = 1, \quad U_{t+u} = U_t U_u, \quad U_t^* = U_{-t}.$$

Define

$$\alpha_t(a) := U_t a U_{-t}, \quad a \in B(\mathcal{H}_S).$$

A direct calculation on basis vectors gives

$$\alpha_t(V_m) = m^{it} V_m. \quad (4)$$

Proposition 1. *For each $t \in \mathbb{R}$, the automorphism α_t preserves $\mathcal{T}_S$, and its restriction*

$$\alpha_t : \mathcal{T}_S \xrightarrow{\sim} \mathcal{T}_S$$

is a $$ -automorphism. The family $(\alpha_t)_{t \in \mathbb{R}}$ satisfies*

$$\alpha_{t+u} = \alpha_t \circ \alpha_u, \quad \alpha_0 = \text{id},$$

and is point-norm continuous. Hence $(\mathcal{T}_S, \alpha)$ is a C^ -dynamical system.*

Proof. Equation (4) shows that every prime generator V_p is sent to the scalar multiple $p^{it} V_p$. Since α_t is a $*$ -automorphism of $B(\mathcal{H}_S)$, it follows that the algebraic $*$ -algebra generated by the V_p 's is invariant. Conjugation by a unitary is isometric, hence norm-continuous, so α_t preserves its norm closure $\mathcal{T}_S$. Applying the same argument to α_{-t} gives equality rather than merely inclusion.

The group law follows from $U_{t+u} = U_t U_u$. For point-norm continuity, first note that

$$\alpha_t(V_m V_n^*) = m^{it} n^{-it} V_m V_n^* = \left(\frac{m}{n}\right)^{it} V_m V_n^*,$$

so $t \mapsto \alpha_t(a)$ is norm-continuous for every element a in the algebraic span of the monomials. Given arbitrary $a \in \mathcal{T}_S$, choose such a finite linear combination b with $\|a - b\| < \varepsilon$. Since every α_t is isometric,

$$\|\alpha_t(a) - \alpha_s(a)\| \leq 2\|a - b\| + \|\alpha_t(b) - \alpha_s(b)\|.$$

The last term tends to zero as $t \rightarrow s$, proving point-norm continuity. $\square$

The self-adjoint generator of this unitary group is the positive diagonal operator

$$H_S e_n = (\log n) e_n$$

with natural domain

$$\text{Dom}(H_S) = \left\{ \xi = (\xi_n)_{n \in \Lambda_S} \in \ell^2(\Lambda_S) : \sum_{n \in \Lambda_S} (\log n)^2 |\xi_n|^2 < \infty \right\}.$$

Thus $U_t = e^{itH_S}$. This Hamiltonian description is useful for Gibbs weights and traces, but the bounded unitaries (3) provide the cleaner definition of the dynamics itself. The logarithmic energy assignment is the same arithmetic scaling mechanism that underlies the time evolution in the Bost–Connes system [1], here restricted to the finite semigroup Λ_S .

4 The finite Euler product as a trace partition function

Lemma 1. *For every real $\beta > 0$, the operator $e^{-\beta H_S}$ is trace class and*

$$Z_S^{\text{fin}}(\beta) := \text{Tr}(e^{-\beta H_S}) = \sum_{n \in \Lambda_S} n^{-\beta} = \prod_{p \in S} (1 - p^{-\beta})^{-1}. \quad (5)$$

In particular, the formula holds for every real $\beta > 1$. Moreover, the right-hand side is holomorphic for $\Re(\beta) > 0$.

Proof. Since $H_S e_n = (\log n) e_n$, we have

$$e^{-\beta H_S} e_n = n^{-\beta} e_n.$$

Therefore

$$\text{Tr}(e^{-\beta H_S}) = \sum_{n \in \Lambda_S} n^{-\beta}.$$

By unique factorization inside the finite semigroup Λ_S , every $n \in \Lambda_S$ has a unique expression

$$n = \prod_{p \in S} p^{k_p}, \quad k_p \in \mathbb{N}.$$

Hence, for $\beta > 0$,

$$\sum_{n \in \Lambda_S} n^{-\beta} = \sum_{(k_p) \in \prod_{p \in S} \mathbb{N}} \prod_{p \in S} p^{-\beta k_p} = \prod_{p \in S} \sum_{k=0}^{\infty} p^{-\beta k} = \prod_{p \in S} (1 - p^{-\beta})^{-1}.$$

The finite product is holomorphic wherever each geometric series converges, namely for $\Re(\beta) > 0$. If $S \neq \emptyset$, the endpoint $\beta = 0$ is sharp: then $n^{-\beta} = 1$ for every $n \in \Lambda_S$, and the infinite sum diverges. If $S = \emptyset$, then $\Lambda_S = \{1\}$ and the partition function is identically 1. This proves the claim. $\square$

5 Existence and uniqueness of the finite-prime KMS state

For $\beta > 0$, it is convenient to write the positive diagonal Gibbs operator as

$$D_{\beta,S} := e^{-\beta H_S} = \sum_{n \in \Lambda_S} n^{-\beta} |e_n\rangle \langle e_n|, \quad (6)$$

where the series converges in trace norm (and hence in operator norm). Its trace is exactly the finite partition function,

$$\mathrm{Tr}(D_{\beta,S}) = Z_S^{\mathrm{fin}}(\beta).$$

Thus the normalized coefficients

$$\mu_{\beta,S}(n) := \frac{n^{-\beta}}{Z_S^{\mathrm{fin}}(\beta)}, \quad n \in \Lambda_S, \quad (7)$$

form a probability distribution on Λ_S :

$$\mu_{\beta,S}(n) \geq 0, \quad \sum_{n \in \Lambda_S} \mu_{\beta,S}(n) = 1.$$

Define

$$\varphi_{\beta,S}(a) := \frac{\mathrm{Tr}(D_{\beta,S}a)}{Z_S^{\mathrm{fin}}(\beta)}, \quad a \in \mathcal{T}_S. \quad (8)$$

This is a state because $D_{\beta,S}$ is positive trace class and its trace is $Z_S^{\mathrm{fin}}(\beta)$. Equivalently, the Gibbs state is the weighted basis-diagonal expectation

$$\varphi_{\beta,S}(a) = \sum_{n \in \Lambda_S} \mu_{\beta,S}(n) \langle e_n, a e_n \rangle. \quad (9)$$

In particular, one may regard $\varphi_{\beta,S}$ as averaging the diagonal matrix coefficient of a against the Gibbs probability distribution on the arithmetic states $n \in \Lambda_S$.

Theorem 1. *For every real $\beta > 0$, the state $\varphi_{\beta,S}$ is the unique KMS_β state on the finite-prime Toeplitz system $(\mathcal{T}_S, \alpha)$. On the dense $*$ -subalgebra spanned by $V_m V_n^*$, it is given by*

$$\varphi_{\beta,S}(V_m V_n^*) = \begin{cases} m^{-\beta}, & m = n, \\ 0, & m \neq n. \end{cases} \quad (10)$$

Proof. First we verify that $\varphi_{\beta,S}$ has the values claimed in (10). For $k \in \Lambda_S$,

$$\langle e_k, V_m V_n^* e_k \rangle$$

can be nonzero only if n divides k inside Λ_S and $mk/n = k$. The latter equality forces $m = n$. Therefore, if $m \neq n$,

$$\mathrm{Tr}(e^{-\beta H_S} V_m V_n^*) = 0.$$

If $m = n$, then $V_m V_m^*$ is the projection onto the closed span of $\{e_{mr} : r \in \Lambda_S\}$. Hence

$$\mathrm{Tr}(e^{-\beta H_S} V_m V_m^*) = \sum_{r \in \Lambda_S} (mr)^{-\beta} = m^{-\beta} Z_S^{\mathrm{fin}}(\beta).$$

Dividing by $Z_S^{\mathrm{fin}}(\beta)$ gives (10).

The Gibbs state (8) satisfies the KMS_β condition. To see this directly on the dense analytic $*$ -subalgebra spanned by the monomials, let

$$A = V_m V_n^*.$$

Then A is entire analytic and

$$\alpha_{i\beta}(A) = (m/n)^{-\beta} A.$$

Writing $\rho_\beta := e^{-\beta H_S}$, the implementation of the dynamics gives

$$\rho_\beta A = \alpha_{i\beta}(A) \rho_\beta.$$

Hence, for analytic monomials A, B , the operator $\rho_\beta A$ is trace class and

$$\text{Tr}(\rho_\beta AB) = \text{Tr}(B \rho_\beta A) = \text{Tr}(B \alpha_{i\beta}(A) \rho_\beta) = \text{Tr}(\rho_\beta B \alpha_{i\beta}(A)).$$

After division by $Z_S^{\text{fin}}(\beta)$, this is precisely

$$\varphi_{\beta,S}(AB) = \varphi_{\beta,S}(B \alpha_{i\beta}(A)).$$

By linearity the same identity holds on the analytic $*$ -algebra spanned by the monomials, and the usual density argument gives the KMS_β condition on $\mathcal{T}_S$.

It remains to prove uniqueness. Let ψ be any KMS_β state on $(\mathcal{T}_S, \alpha)$. Every KMS state is invariant under the dynamics, so

$$\psi(\alpha_t(V_m V_n^*)) = \psi(V_m V_n^*)$$

for all $t \in \mathbb{R}$. But by (4),

$$\alpha_t(V_m V_n^*) = (m/n)^{it} V_m V_n^*.$$

If $m \neq n$, then there exists t such that $(m/n)^{it} \neq 1$, and hence

$$\psi(V_m V_n^*) = 0.$$

If $m = n$, use the KMS condition with $a = V_m$ and $b = V_m^*$. Since

$$\alpha_{i\beta}(V_m) = m^{-\beta} V_m,$$

we obtain

$$\psi(V_m V_m^*) = \psi(V_m^* \alpha_{i\beta}(V_m)) = m^{-\beta} \psi(V_m^* V_m) = m^{-\beta} \psi(1) = m^{-\beta}.$$

Thus ψ agrees with $\varphi_{\beta,S}$ on every monomial $V_m V_n^*$. Since their linear span is dense in $\mathcal{T}_S$, $\psi = \varphi_{\beta,S}$. This proves uniqueness. $\square$

6 The archimedean Gamma factor

The finite-prime KMS theorem accounts for the finite Euler product. The factor

$$\pi^{-s/2} \Gamma\left(\frac{s}{2}\right)$$

is the standard archimedean Mellin factor. Its appearance from the Gaussian is classical and is central to the adelic treatment of zeta and L -functions; see Tate [6].

Let

$$d\mu_\infty(x) := 2e^{-\pi x^2} \frac{dx}{x}, \quad x \in (0, \infty).$$

For real $\beta > 0$, define

$$Z_\infty(\beta) := \int_0^\infty x^\beta d\mu_\infty(x) = 2 \int_0^\infty x^{\beta-1} e^{-\pi x^2} dx. \quad (11)$$

Lemma 2. For $\Re(s) > 0$,

$$2 \int_0^\infty x^{s-1} e^{-\pi x^2} dx = \pi^{-s/2} \Gamma\left(\frac{s}{2}\right). \quad (12)$$

In particular,

$$Z_\infty(\beta) = \pi^{-\beta/2} \Gamma\left(\frac{\beta}{2}\right)$$

for every real $\beta > 0$.

Proof. Using the substitution $u = \pi x^2$, so that

$$x = (u/\pi)^{1/2}, \quad \frac{dx}{x} = \frac{1}{2} \frac{du}{u},$$

we compute

$$2 \int_0^\infty x^s e^{-\pi x^2} \frac{dx}{x} = 2 \int_0^\infty \left(\frac{u}{\pi}\right)^{s/2} e^{-u} \frac{1}{2} \frac{du}{u} = \pi^{-s/2} \int_0^\infty u^{s/2-1} e^{-u} du.$$

The last integral is $\Gamma(s/2)$, valid for $\Re(s) > 0$. This proves the formula. $\square$

The corresponding normalized archimedean probability measure is

$$d\nu_{\beta,\infty}(x) := \frac{x^\beta}{Z_\infty(\beta)} d\mu_\infty(x) = \frac{2x^{\beta-1} e^{-\pi x^2}}{\pi^{-\beta/2} \Gamma(\beta/2)} dx. \quad (13)$$

This is a genuine Radon probability measure on $(0, \infty)$ for every real $\beta > 0$.

7 The combined finite-level state and partition function

Define the product algebra using the minimal (spatial) tensor product,

$$\mathcal{A}_S := \mathcal{T}_S \otimes_{\min} C_0((0, \infty)),$$

with dynamics

$$\sigma_t := \alpha_t \otimes \text{id}.$$

The algebra $C_0((0, \infty))$, and hence $\mathcal{A}_S$, is non-unital but σ -unital. We use the standard non-unital formulation of the KMS condition: the KMS identity is checked on a dense algebra of entire analytic elements and then extended to the C^* -algebra by continuity, equivalently by applying the usual approximate-unit argument; see, for example, [2].

For real $\beta > 0$, define a state on elementary tensors by

$$\omega_{\beta,S}(a \otimes f) := \varphi_{\beta,S}(a) \int_0^\infty f(x) d\nu_{\beta,\infty}(x), \quad (14)$$

and extend linearly and continuously to $\mathcal{A}_S$.

Proposition 2. For every real $\beta > 0$, the state $\omega_{\beta,S}$ is a KMS_β state for $(\mathcal{A}_S, \sigma)$. More generally, every KMS_β state on $(\mathcal{A}_S, \sigma)$ is of the form $\varphi_{\beta,S} \otimes \rho$, where ρ is a state on $C_0((0, \infty))$. Consequently, among KMS states whose archimedean marginal is the probability measure $\nu_{\beta,\infty}$, $\omega_{\beta,S}$ is unique. Its normalizing factor is

$$Z_S(\beta) = Z_S^{\text{fn}}(\beta) Z_\infty(\beta) = \left(\prod_{p \in S} (1 - p^{-\beta})^{-1} \right) \pi^{-\beta/2} \Gamma\left(\frac{\beta}{2}\right). \quad (15)$$

Proof. The state $\varphi_{\beta,S}$ is KMS_β for $(\mathcal{T}_S, \alpha)$ by Theorem 1. The state $f \mapsto \int f d\nu_{\beta,\infty}$ is invariant under the trivial dynamics on $C_0((0, \infty))$. Therefore their tensor product is KMS_β for $\alpha \otimes \text{id}$. Explicitly, for analytic elementary tensors,

$$\omega_{\beta,S}((a \otimes f)\sigma_{i\beta}(b \otimes g)) = \varphi_{\beta,S}(a\alpha_{i\beta}(b)) \int fg d\nu_{\beta,\infty},$$

while

$$\omega_{\beta,S}((b \otimes g)(a \otimes f)) = \varphi_{\beta,S}(ba) \int gf d\nu_{\beta,\infty}.$$

The two quantities are equal by the finite-prime KMS condition and commutativity of the archimedean factor. Density gives the result for $\mathcal{A}_S$.

The formula for $Z_S(\beta)$ follows immediately by multiplying Lemma 1 and Lemma 2.

It remains to justify the uniqueness statement in the non-unital tensor product. Let ω be any KMS_β state on $(\mathcal{A}_S, \sigma)$. For $0 \leq f \in C_0((0, \infty))$, define a positive functional on $\mathcal{T}_S$ by

$$\psi_f(a) := \omega(a \otimes f).$$

If $\psi_f(1) > 0$, then the normalized functional $\psi_f/\psi_f(1)$ is a KMS_β state on $(\mathcal{T}_S, \alpha)$. Indeed, applying the KMS identity in $\mathcal{A}_S$ to $a \otimes f^{1/2}$ and $b \otimes f^{1/2}$ gives

$$\psi_f(a\alpha_{i\beta}(b)) = \psi_f(ba)$$

for analytic $a, b \in \mathcal{T}_S$. By Theorem 1, $\psi_f = \psi_f(1)\varphi_{\beta,S}$. If $\psi_f(1) = 0$, positivity gives $\psi_f = 0$, so the same conclusion holds.

Now set

$$\rho(f) := \omega(1 \otimes f), \quad f \in C_0((0, \infty)).$$

Then ρ is a positive functional. If (u_j) is a positive contractive approximate unit for $C_0((0, \infty))$, then $1 \otimes u_j$ is an approximate unit for $\mathcal{A}_S$, so $\rho(u_j) \rightarrow \|\omega\| = 1$. Hence ρ is a state. The preceding paragraph shows

$$\omega(a \otimes f) = \varphi_{\beta,S}(a)\rho(f)$$

for all $a \in \mathcal{T}_S$ and all positive $f \in C_0((0, \infty))$, and therefore for all f by linearity. Thus $\omega = \varphi_{\beta,S} \otimes \rho$. Taking $\rho(f) = \int f d\nu_{\beta,\infty}$ gives the particular state (14), and it is the only KMS state with that archimedean marginal. $\square$

Remark 1 (Can the archimedean factor be made operator-algebraic?). *Yes, but not in quite the same trace-class way as the finite-prime Toeplitz factor. Let*

$$\Phi_\infty(f) := 2 \int_0^\infty f(x) e^{-\pi x^2} \frac{dx}{x}$$

be the natural Gaussian Mellin weight on $C_0((0, \infty))$. If M_x denotes multiplication by x , then formally

$$Z_\infty(\beta) = \Phi_\infty(M_x^\beta)$$

for $\beta > 0$. This is a legitimate commutative operator-algebraic normalization by a semifinite weight, but it is not a discrete trace partition function $\text{Tr}(e^{-\beta H})$ on ℓ^2 unless one adds an additional discretization. In particular, nothing in the present construction identifies the product normalization $Z_S^{\text{fin}}(\beta)Z_\infty(\beta)$ with $\text{Tr}(e^{-\beta H})$ for a single trace-class heat operator on a Hilbert-space representation of $\mathcal{A}_S$. For this reason the paper keeps the finite-prime trace-class Gibbs normalization and the archimedean Mellin normalization conceptually separate.

8 Relation to the Bost–Connes system and scope

The full Bost–Connes system contains substantially more structure than the finite-prime model considered here. In particular, its arithmetic symmetry, its low-temperature extremal KMS states, and the spontaneous symmetry breaking described in [1] are not present in the construction above. We have retained only the multiplicative semigroup generated by a finite set of primes and the logarithmic time evolution that turns multiplication into additive energy.

Nevertheless, the finite model isolates one important mechanism of the arithmetic quantum statistical picture. Each prime contributes a local geometric partition factor

$$Z_p(\beta) = (1 - p^{-\beta})^{-1},$$

and unique factorization makes the finite-prime partition function the product of these independent local factors:

$$Z_S^{\text{fin}}(\beta) = \prod_{p \in S} Z_p(\beta).$$

At fixed finite S , this expression is holomorphic for $\Re(s) > 0$. If S increases through the finite subsets of all rational primes, the Euler products converge for $\Re(s) > 1$ to

$$\zeta(s) = \prod_p (1 - p^{-s})^{-1}.$$

Thus the familiar half-plane $\Re(s) > 1$ is a global phenomenon: it is not needed at any fixed finite level.

The archimedean factor should also be interpreted carefully. The finite prime contribution is the trace of a trace-class Gibbs operator on $\ell^2(\Lambda_S)$. By contrast,

$$\pi^{-s/2} \Gamma(s/2)$$

arises here from a Gaussian Mellin transform, as in the classical local analysis of zeta functions [6]. Their product is therefore a natural finite-place-plus-archimedean normalization, but the present paper does not identify it with the trace of one heat operator on a single Hilbert-space representation. Keeping this distinction explicit prevents the analogy with statistical mechanics from being stronger than the construction justifies.

9 Conclusion

For a finite set S of rational primes, we have constructed an explicit C^* -dynamical system

$$(\mathcal{T}_S, \alpha), \quad \mathcal{T}_S = C^*(1, V_p : p \in S), \quad \alpha_t(V_m) = m^{it} V_m,$$

and proved that for every real $\beta > 0$ it has a unique KMS_β state. On the dense monomial algebra,

$$\varphi_{\beta,S}(V_m V_n^*) = \delta_{m,n} m^{-\beta},$$

and its Gibbs partition function is

$$Z_S^{\text{fin}}(\beta) = \sum_{n \in \Lambda_S} n^{-\beta} = \prod_{p \in S} (1 - p^{-\beta})^{-1}.$$

The real place contributes the classical Mellin–Gaussian factor

$$Z_{\infty}(\beta) = \pi^{-\beta/2} \Gamma\left(\frac{\beta}{2}\right),$$

so the combined normalization is

$$Z_S(\beta) = \pi^{-\beta/2} \Gamma\left(\frac{\beta}{2}\right) \prod_{p \in S} (1 - p^{-\beta})^{-1}.$$

The construction is deliberately elementary: finite unique factorization, unilateral shifts, a diagonal logarithmic dynamics, Gibbs weights, and the KMS condition suffice. In this form the finite Euler factors appear not as a formal analogy but as the literal partition function of a concrete Toeplitz dynamical system.

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